

Hafiz Adjei

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EDUCATION

Carnegie Mellon University Africa Kigali, Rwanda **Aug. 2025 - May 2026**

Master of Science in Electrical and Computer Engineering

- Concentrations: Wireless Communications & Network Applications, Deep Learning, IoT, Applied Machine Learning
- Cumulative GPA: 3.69/4.0

Ashesi University Berekusu, Ghana

Sep. 2020 - Aug. 2024

Bachelor of Science in Electrical and Electronic Engineering

- Honors: Dean's List (2021-2024), MasterCard Foundation Scholar (full merit scholarship)
- Cumulative GPA: 3.53/4.0

WORK EXPERIENCE/INTERNSHIPS

KCRC / CMU Africa Kigali, Rwanda **May 2026 - Present**

Research Associate

- Reverse-engineered a proprietary 5V TTL UART protocol from legacy documentation to integrate battery modules with a modern monitoring system
- Developed a full-stack BMS monitoring system comprising a Python serial communication library, Flask REST API, and a real-time web dashboard
- Implemented multi-module daisy-chain addressing to poll 3 battery modules simultaneously with validated packets

Carnegie Mellon University Africa Kigali, Rwanda

Sep. 2025 - May 2026

Student IT Support

- Delivered real-time AV technical support during lectures and large-scale university events, maintaining near-zero downtime
- Assisted network and server engineers with LAN administration, fault diagnosis, and data-center operations

Ashesi University Berekusu, Ghana

Sep. 2024 - May 2025

Teaching Assistant

- Organized and delivered laboratory sessions and tutorials for undergraduate engineering courses, directly supporting improved student performance
- Collaborated with course instructors to identify academic and behavioral challenges and propose targeted interventions

Northern Electricity Distribution Company Tamale, Ghana

Jul. 2023 - Aug. 2023

Electrical Technician Intern

- Executed service-drop installations and electricity meter deployments for 500+ customers, expanding grid access across northern Ghana
- Verified monthly billing data against meter-reader records to ensure billing accuracy and reduce revenue leakage

PROJECTS

Smart City Parking System CMU Africa **Nov. 2025 - Dec. 2025**

- Designed a low-cost smart parking system integrating inductive sensing, ultrasonic validation, and edge computing for high-accuracy vehicle detection
- Built a web-based dashboard for live slot-occupancy monitoring, real-time analytics, and local session tracking

Autonomous Ground Robot for Greenhouse Monitoring Ashesi University

Jan. 2024 - Oct. 2024

- Designed and deployed a real-time environmental monitoring system enabling data-driven crop optimisation for greenhouse farmers
- Engineered autonomous navigation and path-planning algorithms for the robot; built PHP/HTML dashboards for remote monitoring

LEADERSHIP EXPERIENCE

Ashesi Students Council Berekusu, Ghana

Mar. 2023 - Apr. 2024

Academic Committee Member

- Supervised peer-tutoring programme covering mathematics and computer programming for underclassmen
- Spearheaded academic integrity awareness campaigns to deter plagiarism campus-wide

CORE COMPETENCIES AND SKILLS

- Data Analytics & Programming: Python, C, JavaScript, PHP, HTML, Data Analytics, TensorFlow, Deep Learning
- Consulting Tools: Microsoft Excel (Advanced), Microsoft PowerPoint (Advanced), Structured Reporting
- Engineering & IoT: Embedded Systems, ARM Assembly, ESP32, Arduino, MQTT, Bluetooth, Technical Troubleshooting

CERTIFICATIONS

- Harvard Health Systems Innovation Lab Hackathon: Certificate in Building High-Value Health Systems Leveraging AI (Apr. 2025)